Questions

Q1.

Given that

$$y = \frac{5x^2 - 10x + 9}{(x-1)^2} \qquad x \neq 1$$

show that $\frac{dy}{dx} = \frac{k}{(x-1)^3}$, where k is a constant to be found.

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(Total for question = 6 marks)

(Q05 WMA13/01, Specimen papers)

Q2.

The curve C has equation

$$y = (3x - 2)^6$$

(a) Find $\frac{dy}{dx}$

(2)

Given that the point $P\left(\frac{1}{3}, 1\right)$ lies on C,

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(b) find the equation of the normal to C at P. Write your answer in the form $ax + by + c = 0$ where a, b and c are integers to be found.

(4)

(Total for question = 6 marks)

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(Q01 WMA13/01, June 2022)

Q3.

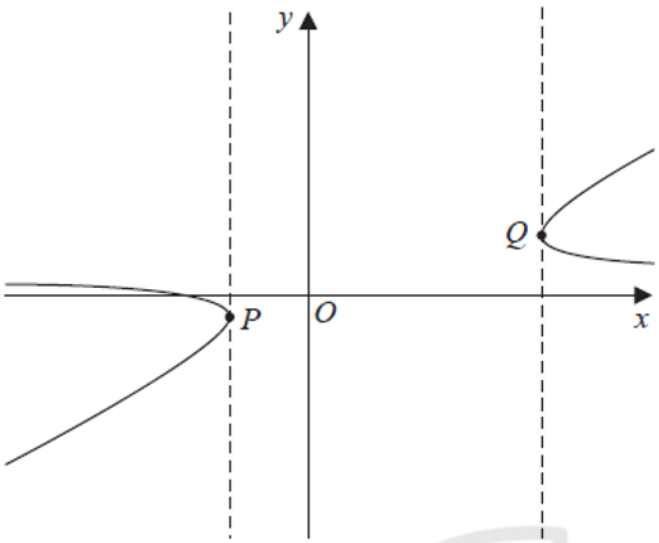


Figure 4

Figure 4 shows a sketch of the curve with equation

$$x = \frac{2y^2 + 6}{3y - 3}$$

(a) Find $\frac{dx}{dy}$ giving your answer as a fully simplified fraction.

(4)

The tangents at points P and Q on the curve are parallel to the y-axis, as shown in Figure 4.

(b) Use the answer to part (a) to find the equations of these two tangents.

(4)

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(Total for question = 8 marks)

(Q10 WMA13/01, June 2023)

Q4.

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The function f is defined by

$$f(x) = \frac{5x}{x^2 + 7x + 12} + \frac{5x}{x + 4} \qquad x > 0$$

(a) Show that $f(x) = \frac{5x}{x + 3}$

(3)

(b) Find f^{-1}

(3)

(c) (i) Find, in simplest form, $f'(x)$.

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(ii) Hence, state whether f is an increasing or a decreasing function, giving a reason for your answer.

(3)

(Total for question = 9 marks)

(**Q01 WMA13/01, Oct 2021**)

Q5.

The function f is defined by

$$f(x) = \frac{5x - 3}{x - 4} \qquad x > 4$$

(a) Show, by using calculus, that f is a decreasing function.

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(3)

(b) Find f^{-1}

(3)

(c) (i) Show that $ff(x) = \frac{ax + b}{x + c}$ where a, b and c are constants to be found.

(ii) Deduce the range of ff.

(5)

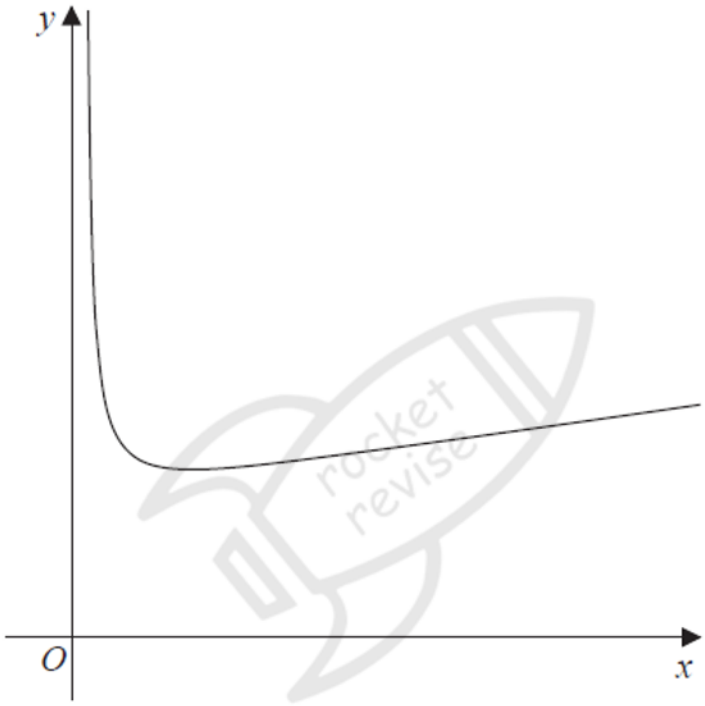
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(Total for question = 11 marks)

(Q06 WMA13/01, Jan 2022)

Q6.



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Figure 1

Figure 1 shows a sketch of a curve with equation $y = f(x)$ where

$$f(x) = \frac{2x+3}{\sqrt{4x-1}} \qquad x > \frac{1}{4}$$

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(a) Find, in simplest form, $f'(x)$.

(4)

(b) Hence find the range of f.

(3)

(Total for question = 7 marks)

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(Q03 WMA13/01, Oct 2020)

Q7.

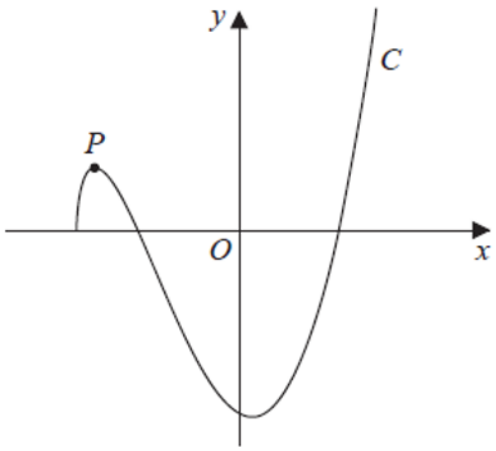


Figure 3

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

The function f is defined by

$$f(x) = 5(x^2 - 2)(4x + 9)^{\frac{1}{2}} \qquad x \geqslant -\frac{9}{4}$$

(a) Show that

$$f'(x) = \frac{k(5x^2 + 9x - 2)}{(4x + 9)^{\frac{1}{2}}}$$

where k is an integer to be found.

(4)

(b) Hence, find the values of x for which $f'(x) = 0$

(1)

Figure 3 shows a sketch of the curve C with equation $y = f(x)$.

The curve has a local maximum at the point P

(c) Find the exact coordinates of P

(2)

The function g is defined by

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$g(x) = 2f(x) + 4$ $\quad -\frac{9}{4} \leq x \leq 0$

(d) Find the range of g

(3)

(Total for question = 10 marks)

(Q06 WMA13/01, June 2022)

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Q8.

(a) Given that

$$\frac{x^4 - x^3 - 10x^2 + 3x - 9}{x^2 - x - 12} \equiv x^2 + P + \frac{Q}{x - 4} \qquad x > -3$$

find the value of the constant P and show that $Q = 5$

(4)

The curve C has equation $y = g(x)$, where

$$g(x) = \frac{x^4 - x^3 - 10x^2 + 3x - 9}{x^2 - x - 12} \qquad -3 < x < 3.5 \qquad x \in \mathbb{R}$$

(b) Find the equation of the tangent to C at the point where $x = 2$
Give your answer in the form $y = mx + c$, where m and c are constants to be found.

(5)

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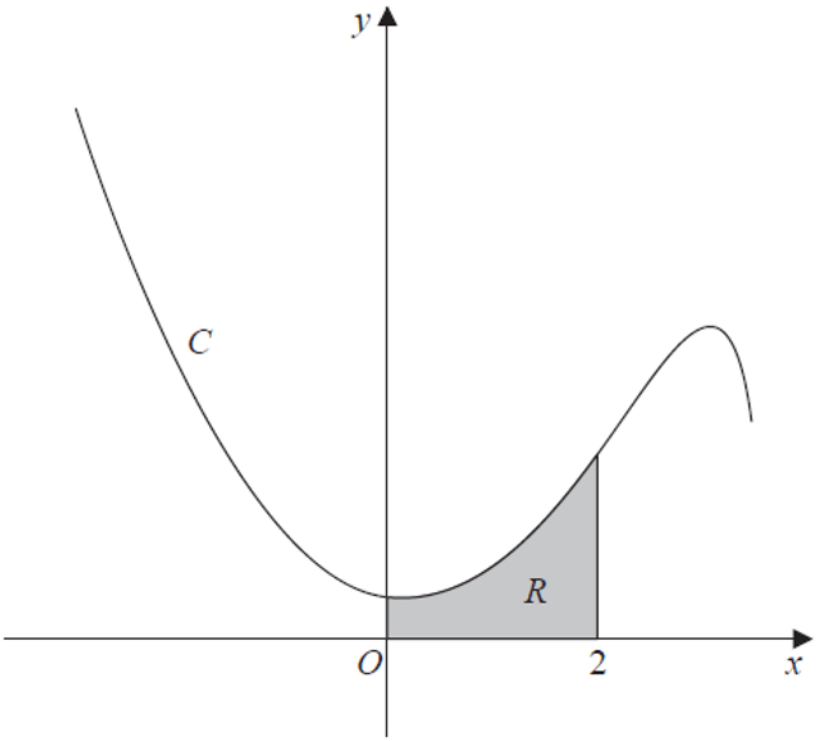


Figure 4

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Figure 4 shows a sketch of the curve C.
The region R, shown shaded in Figure 4, is bounded by C, the y-axis, the x-axis and the line with equation $x = 2$

(c) Find the exact area of R, writing your answer in the form $a + b\ln 2$, where a and b are constants to be found.

(5)

(Total for question = 14 marks)

(Q09 WMA13/01, Oct 2020)

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